

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

The alpaca was domesticated by Indigenous peoples in the Andes about 7,000 years ago. But which wild species did it descend from, the vicuña or guanaco? A research team led by Ruiwen Fan may have solved the mystery, concluding that the alpaca is the domesticated form of the vicuña but that the modern alpaca gets only 64 percent of its genetic material from its wild ancestor. The rest comes from the domesticated llama. The llama, meanwhile, gets 95.5 percent of its genetic material from its own wild ancestor, the guanaco, and the rest from the alpaca. The llama and alpaca apparently interbred widely for only a handful of generations between 400 and 600 years ago. Assuming that the findings of Fan’s team are valid, it can be inferred that _____

Which choice most logically completes the text?

Answer

- A. modern llama populations have a greater degree of genetic diversity, on average, than modern alpaca populations do.
- B. the domestication process of the alpaca may have involved some introduction of genetic material from the llama.
- C. the period of interbreeding resulted in a greater genetic difference between alpacas and their wild ancestors than between llamas and their wild ancestors.
- D. if they were subjected to genetic testing, modern populations of guanacos and vicuñas would likely show traces of ancient interbreeding as well.

Correct Answer: C

Rationale

Choice C is the best answer because it presents the inference that most logically follows from the text’s discussion of the genetic material of alpacas. The text states that modern alpacas get only 64 percent of their genetic material from their wild ancestor (the vicuña), while llamas get 95.5 percent of their genetic material from their wild ancestor (the guanaco). These data imply that relative to llamas, alpacas have experienced a much greater genetic shift away from their wild ancestors (36 percent of their genetic material derives from another source, compared to 4.5 percent of llamas’ genetic material). Moreover, the text establishes that the remainder of alpacas’ genetic material derives from llamas, and the remainder of llamas’ genetic material derives from alpacas, reflecting a period of interbreeding. Taken together, these details support the inference that the period of interbreeding had a more significant impact on the genetic material of alpacas than on that of llamas, thereby reducing the percentage of alpacas’ genetic material that derives from wild ancestors and placing alpacas at a greater remove from these ancestors.

Choice A is incorrect because the text doesn’t provide information about genetic diversity within modern llama and alpaca populations. While the text discusses the percentage of genetic material each domesticated species inherited from its wild ancestor and from interbreeding, it does not address the level of genetic variation within each species. Therefore, it isn’t logical to infer that modern llama populations have a greater degree of genetic diversity than modern alpaca populations do. Choice B is incorrect because it contradicts the chronology presented in the text. The text states that alpacas were domesticated about 7,000 years ago, but the interbreeding with llamas occurred “between 400 and 600 years ago.” The genetic mixing happened thousands of years after domestication, not during it; in addition, the text does not provide details about what the process of domestication might entail. Therefore, it isn’t logical to infer that llama genetic material was introduced during the domestication process of alpacas. Choice D is incorrect because though the text notes that guanacos and vicuñas contributed to the genetic material of alpacas and llamas, it only discusses alpacas’ and llamas’ genetic composition. Therefore, the text does not support the inference that modern populations of guanacos and vicuñas would likely show traces of ancient interbreeding.